# AGILE DEVELOPMENT PROCESS AND DEVOPS

# FAT

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SET-3

Problem statement:

Ensure listing and lending books

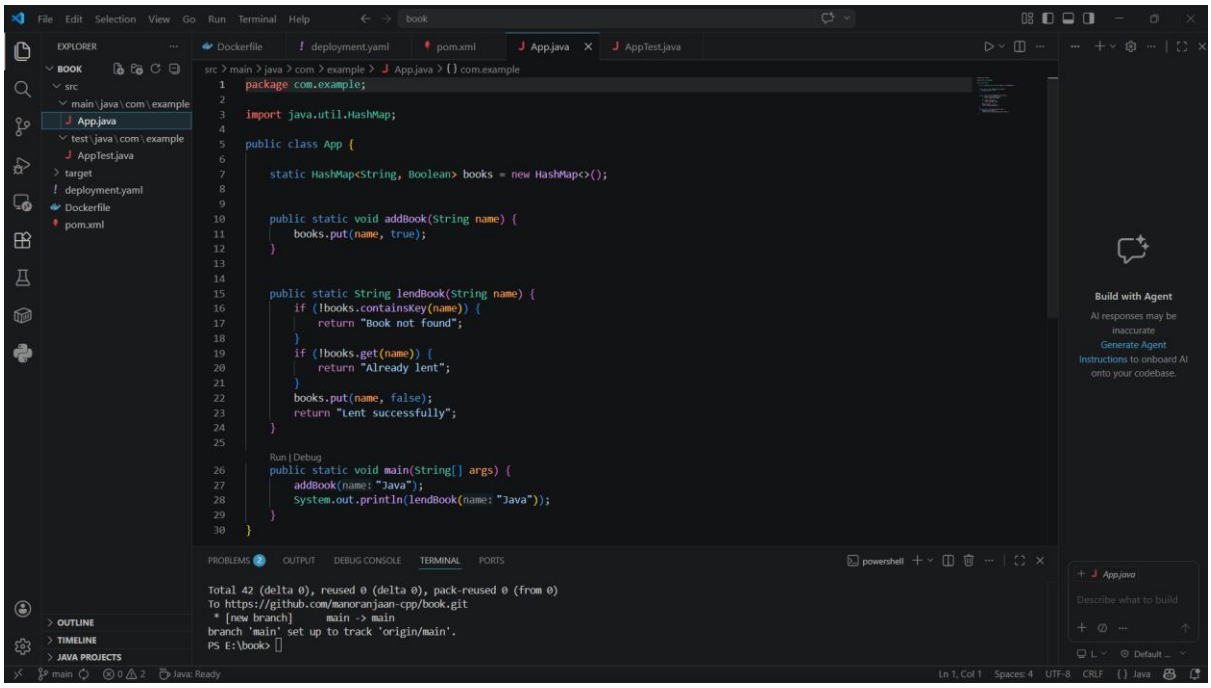
task:

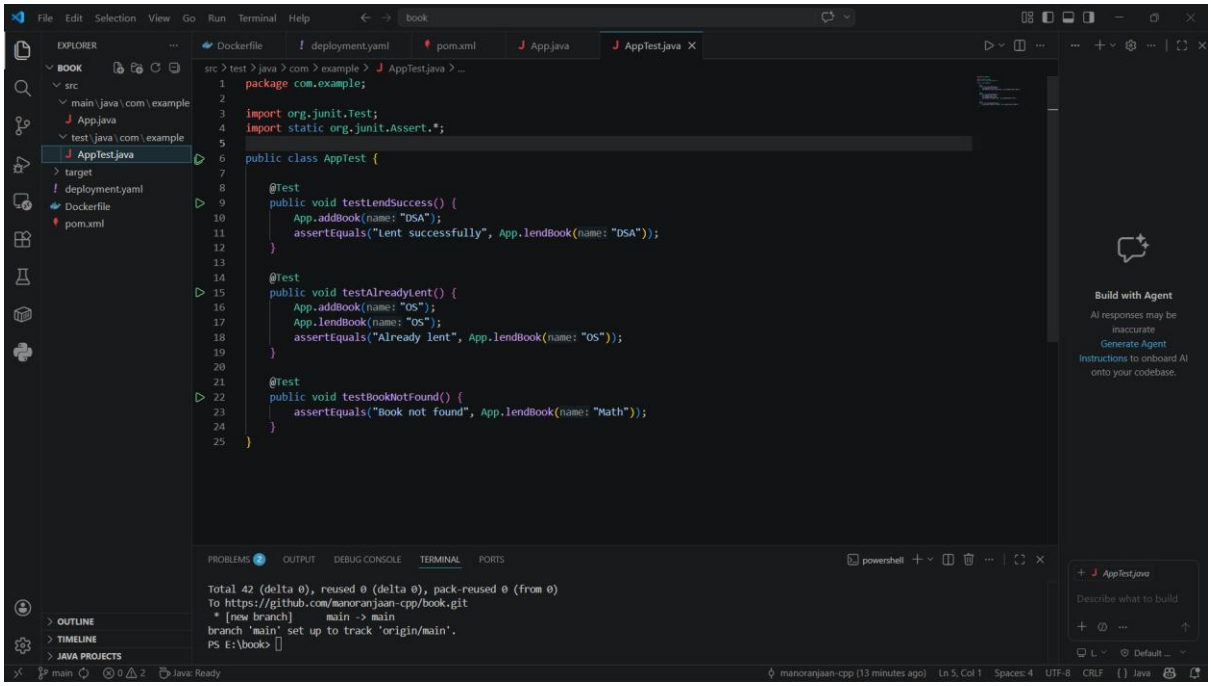
1.develop book lisiting and lending logic

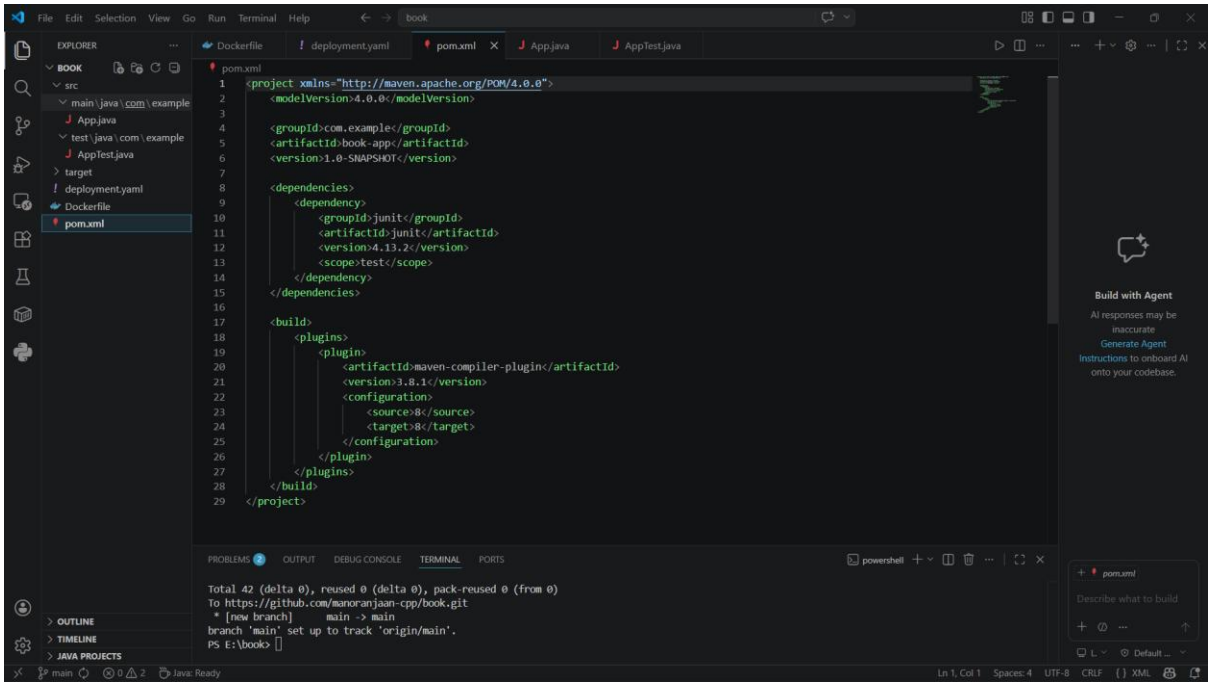
2.write junit test for lending success/failure scenarios

3.implement CI/CD pipeline with automated testing

1.Code for App.java,AppTest.java,pom.xml,deployment.yaml,Dockerfile




File Edit Selection View Go Run Terminal Help
book
EXPLORER
BOOK
src
main \ java \ com \ example
App.java
test \ java \ com \ example
AppTest.java
target
deployment.yaml
Dockerfile
pom.xml
Dockerfile
deployment.yaml
pom.xml
App.java
AppTest.java
src > test > java > com > example > AppTest.java > ...
package com.example;
import org.junit.Test;
import static org.junit.Assert.*;
public class AppTest {
@Test
public void testLendSuccess() {
App.addBook(name: "DSA");
assertEquals("Lent successfully", App.lendBook(name: "DSA"));
}
@Test
public void testAlreadyLent() {
App.addBook(name: "OS");
App.lendBook(name: "OS");
assertEquals("Already lent", App.lendBook(name: "OS"));
}
@Test
public void testBookNotFound() {
assertEquals("Book not found", App.lendBook(name: "Math"));
}
}
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
Total 42 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manoranjaan-cpp/book.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
PS E:\book>
OUTLINE
TIMELINE
JAVA PROJECTS
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
AppTest.java
Describe what to build
main
Java: Ready
manoranjaan-cpp (13 minutes ago)
Ln 5, Col 1
Spaces: 4
UTF-8
CRLF
Java


File Edit Selection View Go Run Terminal Help
book
EXPLORER
BOOK
src
main \ java \ com \ example
App.java
test \ java \ com \ example
AppTest.java
target
deployment.yaml
Dockerfile
pom.xml
Dockerfile
deployment.yaml
pom.xml
App.java
AppTest.java
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>book-app</artifactId>
<version>1.0-SNAPSHOT</version>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>4.13.2</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>8</source>
<target>8</target>
</configuration>
</plugin>
</plugins>
</build>
</project>
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
Total 42 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manoranjaan-cpp/book.git
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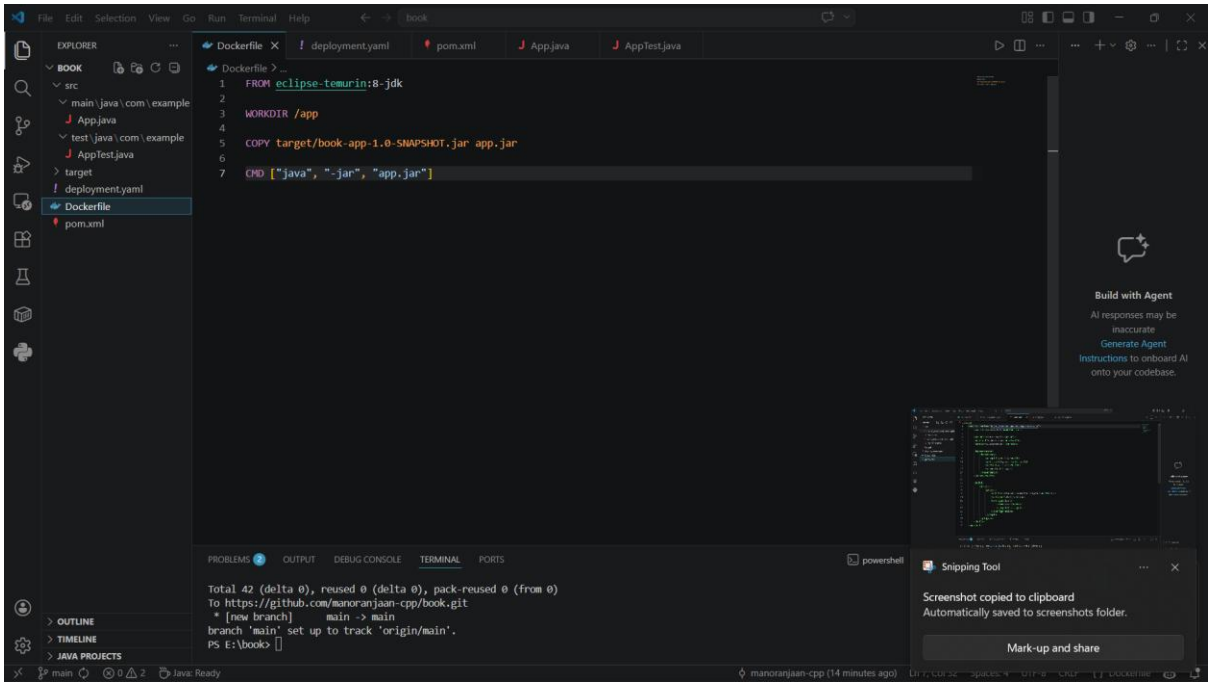
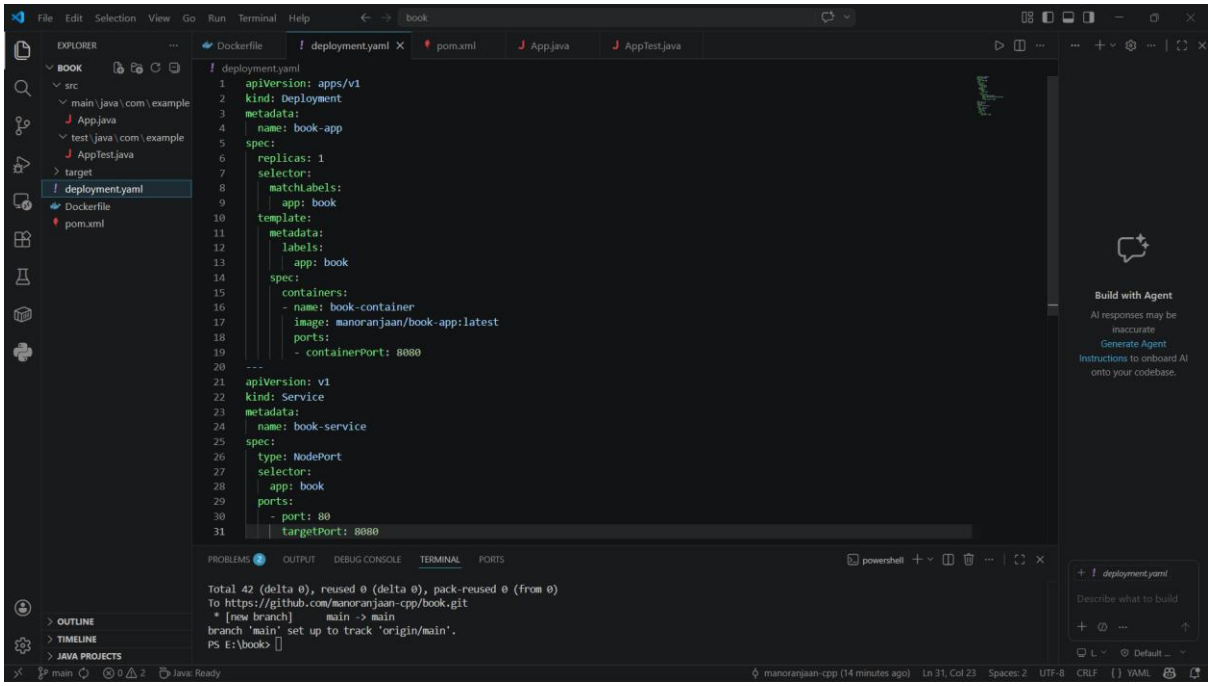
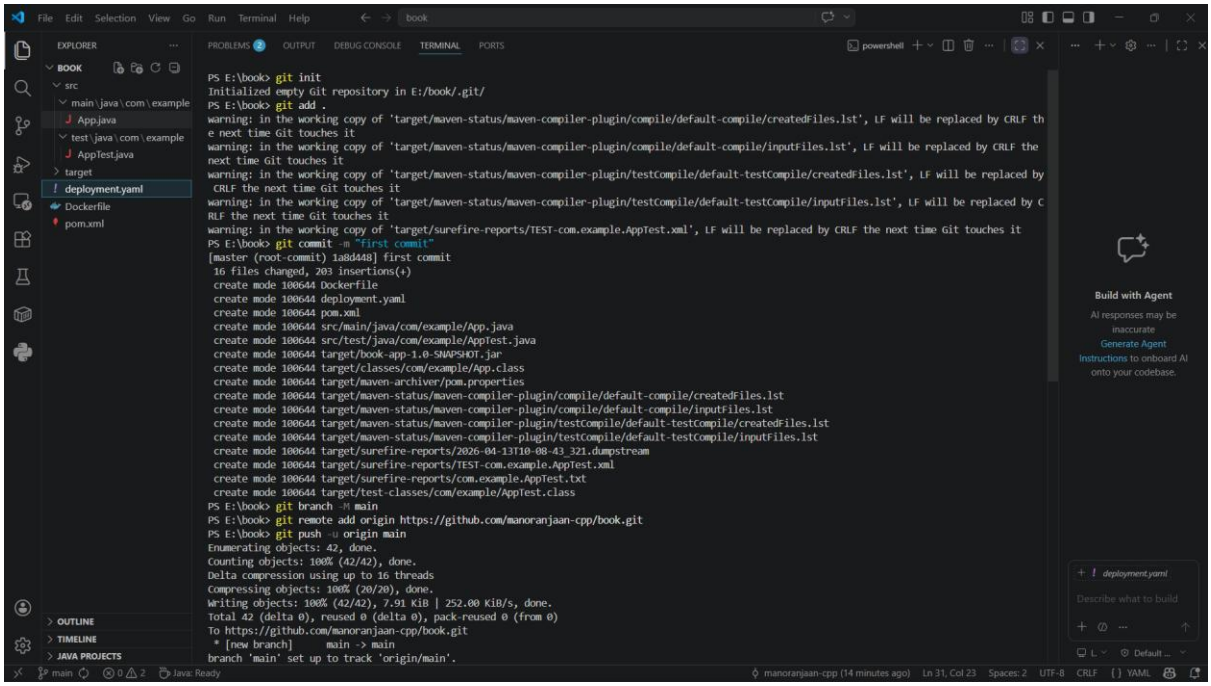
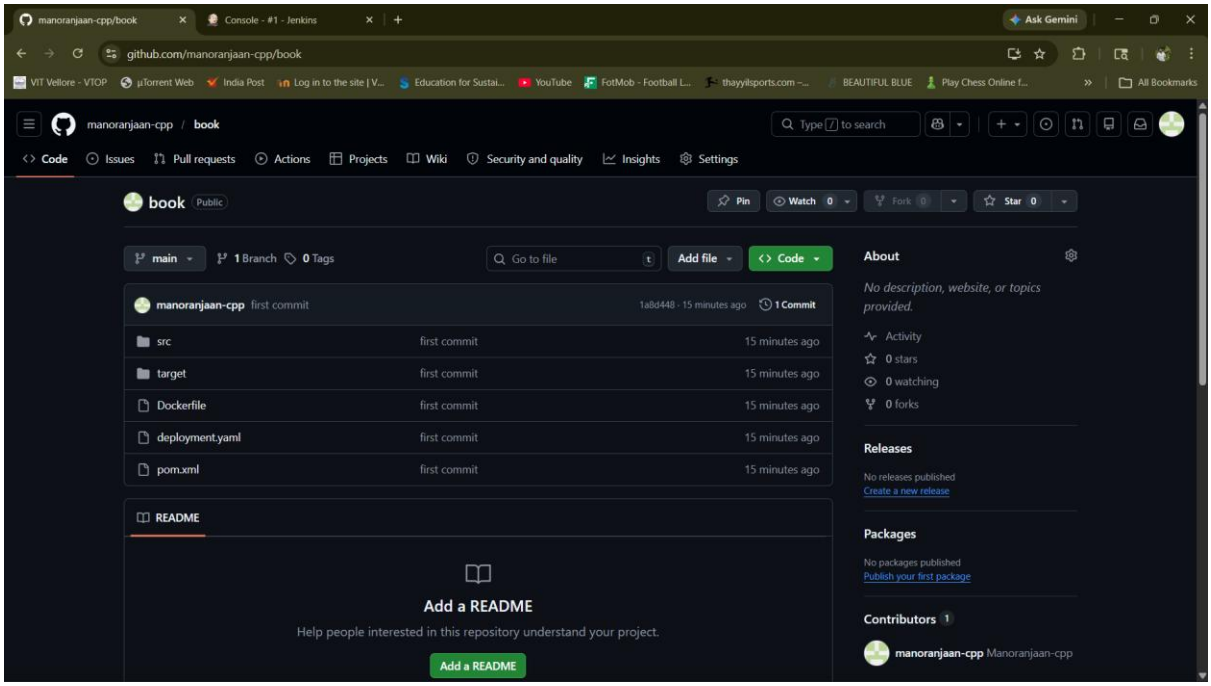
OUTLINE
TIMELINE
JAVA PROJECTS
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
pom.xml
Describe what to build
main
Java: Ready
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
XML




2.Created rep called book in github and pushed all these code




3.builded in local using maven

```
[INFO] --- resources:3.4.0:resources (default-resources) @ book-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory E:\book\src\main\resources
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ book-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to E:\book\target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ book-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory E:\book\src\test\resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ book-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to E:\book\target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ book-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.063 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ book-app ---
[INFO] Building jar: E:\book\target\book-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.339 s
[INFO] Finished at: 2026-04-13T10:08:44+05:30
[INFO] ------------------------------------------------------------------------

E:\book>
```

4.created Jenkins pipeline called book and the pipeline script is :

```
localhost:8081/job/book/1/console
Jenkins / book / #1

418c3b317b95: Waiting
df075017d151: Waiting
418c3b317b95: Mounted from manoranjaan/calc
df075017d151: Waiting
df075017d151: Waiting
df075017d151: Waiting
df075017d151: Mounted from manoranjaan/calc
ac2e62c87387: Pushed
8f8f10646cdb: Pushed
latest: digest: sha256:2ca0e08fd92ed122a3aec87c8036ca051a758fa0aecccd816a4d585a60a23bc3 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\book>cd book   && set KUBECONFIG=C:\Users\manor\.kube\config   && kubectl apply -f deployment.yaml
deployment.apps/book-app created
service/book-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
🎉 PIPELINE SUCCESS
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pipeline script:

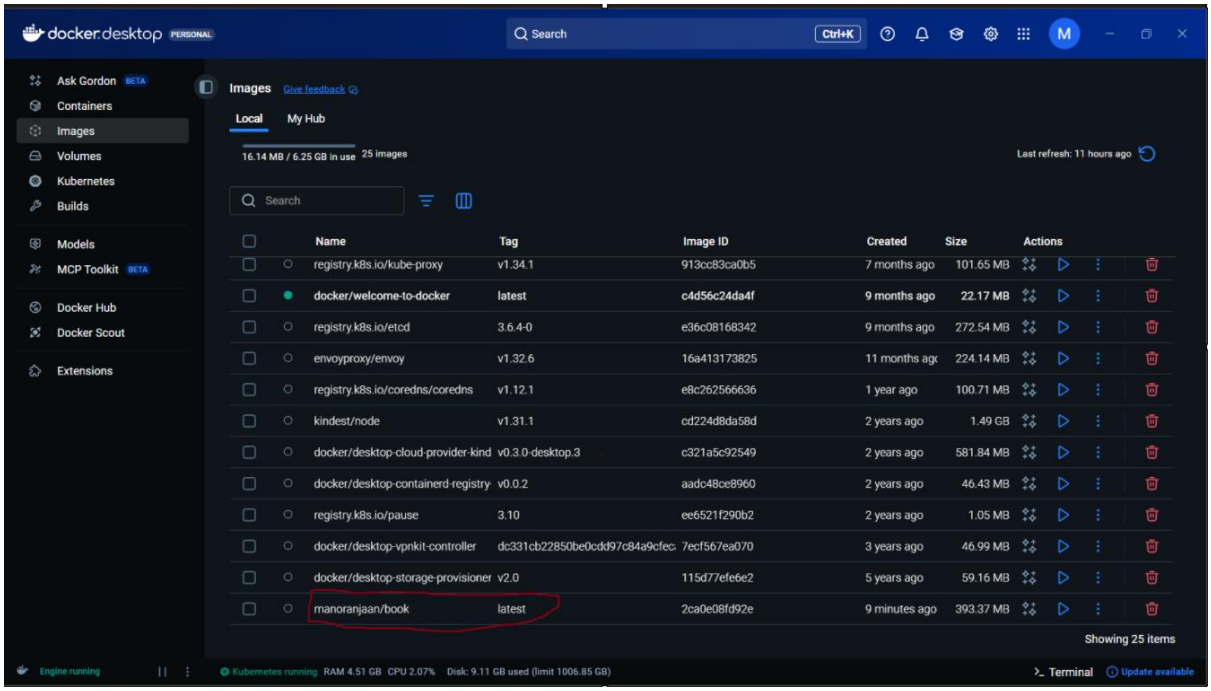
```
pipeline {

   agent any
```

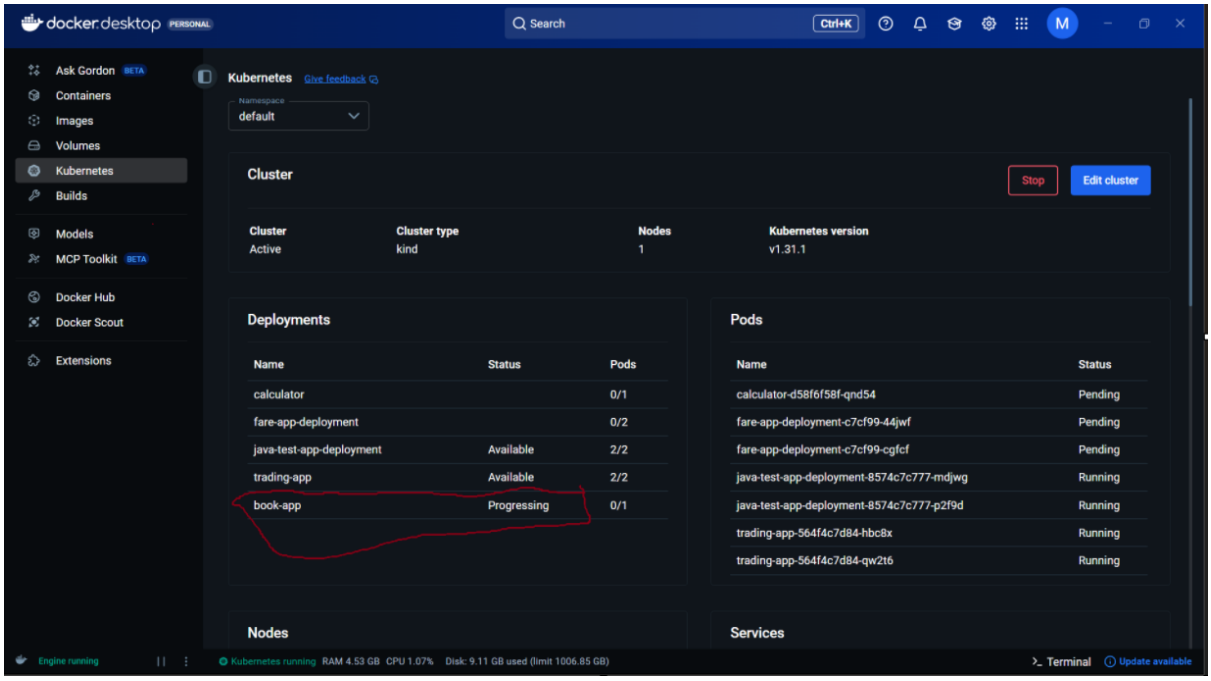
```
environment {
  DOCKER_IMAGE = "manoranjaan/book:latest"
}

stages {

  stage('Clean') {
    steps {
      deleteDir()
    }
  }

  stage('Clone') {
    steps {
      bat 'git clone https://github.com/manoranjaan-cpp/book.git'
    }
  }

  stage('Build & Test') {
    steps {
      bat 'cd book && mvn clean test'
    }
    post {
      always {
        junit 'book/target/surefire-reports/*.xml'
```

```
        }
      }
    }

    stage('Package') {
      steps {
        bat 'cd book && mvn package -DskipTests'
      }
    }

    stage('Docker Build') {
      steps {
        bat 'cd book && docker build -t manoranjaan/book .'
      }
    }

    stage('Docker Push') {
      steps {
        withCredentials([usernamePassword(
          credentialsId: 'dockerhub-creds',
          usernameVariable: 'DOCKER_USER',
          passwordVariable: 'DOCKER_PASS'
        )]) {
          bat 'echo %DOCKER_PASS% | docker login -u %DOCKER_USER%
--password-stdin'
```

```
            bat 'docker push manoranjaan/book'
          }
        }
      }

      stage('Deploy') {
        steps {
          bat 'cd book && set KUBECONFIG=C:\\Users\\manor\\.kube\\config
&& kubectl apply -f deployment.yaml'
        }
      }
    }

    post {
      success {
        echo ' PIPELINE SUCCESS'
      }
      failure {
        echo 'PIPELINE FAILED'
      }
    }
}
```

5.verification of docker image and k8’s in docker desktop




6.verifiying services of k8's in bash(local terminal)

```
E:\book>kubectl get svc
NAME           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
book-service   NodePort    10.96.119.122   <none>        80:30410/TCP   10m
kubernetes     ClusterIP   10.96.0.1       <none>        443/TCP        10h

E:\book>
```